SECTION 4: RADAR & RF SUBSYSTEM REQUIREMENTS

Hierarchical Technical Specification Clauses & Nested Verification Tree

Clause 4.1 Frequency Generation & Synthesizers

4.1.1 The master microwave frequency source shall satisfy the following operational bounds:

(a) Frequency tuning range: 2.0 GHz to 18.0 GHz continuously tunable in 100 Hz increments;

(b) Output power level: +15 dBm minimum leveled power across the entire frequency spectrum;

(c) Spectral purity and phase noise constraints:

(i) Phase noise at 10 kHz offset: <= -115 dBc/Hz across all operational bands;

(ii) Phase noise at 100 kHz offset: <= -128 dBc/Hz at 10 GHz carrier frequency;

(iii) Harmonic spurious suppression: > 45 dBc below carrier at rated maximum output;

Clause 4.2 Wideband Digital Receiver & Signal Channel

4.2.1 High-speed analog-to-digital converter (ADC) and signal channel criteria:

(a) Sampling rate: minimum 2.5 GSPS with 14-bit digitizer resolution;

(b) Analog instantaneous bandwidth: DC to 4.0 GHz (-3 dB cutoff point);

(c) Spurious-Free Dynamic Range (SFDR): >= 72 dBc for input tones up to 1 GHz;

(d) Real-time FPGA processing: Xilinx UltraScale+ FPGA with minimum 8 GB DDR4 buffer memory;

Clause 4.3 Digital Signal Processing (DSP) Algorithms

4.3.1 Embedded DSP firmware functions:

(a) Polyphase filter bank: minimum 1024 channels with >= 60 dB stopband rejection;

(b) Real-time pulse compression: matched filter implementation for chirp waveforms up to 500 MHz;

(c) Latency requirement: overall pipeline processing latency shall not exceed 1.5 microseconds;